Structural Defect Dataset Curation Framework

What are we trying to achieve?

Goal:

- Curate a high-quality structural defect detection dataset
- Improve reliability of AI-based structural inspection models
- Extend the semantic segmentation to instance segmentation

Key idea:

- Better data $\rightarrow$ better model performance

2. Which dataset are we curating?

We use DACL10K dataset

- Large-scale dataset for bridge defect inspection
- Collected from real-world bridge inspections in Germany
- Designed for detection and segmentation tasks

Key statistics:

- $\sim$8,922 images
- 19 object classes (defects + structural components)
- Bounding boxes + pixel-level polygon annotations

DACL10K: Data Characteristics

Real-world dataset properties:

- Captured from real bridge inspections in Germany
- Includes diverse environmental and imaging conditions
- Reflects real engineering inspection scenarios

Challenges in the data:

- Large variation in defect scale and appearance
- Small and irregularly shaped defects (e.g., fine cracks, seepage)

DACL10K: Annotation Format

Two annotation levels:

- Bounding boxes for coarse localization
- Polygon masks for pixel-level segmentation

Key insight:

- Enables both detection and segmentation tasks
- Annotation quality depends strongly on defect visibility

Semantic vs Instance Segmentation

Semantic Segmentation

- Pixel-level class labeling
- No separation between individual defect instances

Instance Segmentation

- Separates each defect instance individually
- Provides finer structural understanding

Goal:

- Move toward instance-level defect understanding

Why Dataset Curation is Needed?

- Not all images are suitable for annotation
- Variability in blur, brightness, contrast, resolution
- Some defects are unclear or ambiguous

Therefore: We must filter and tag data before annotation

Dataset Curation Process

Curation includes:

- Image quality assessment
- Tagging images
- Filtering unsuitable samples
- Selecting annotation-ready data

How do we start dataset curation?

We begin by tagging images using a Three-Stage Decision Framework

1. Image Quality Assessment
2. Defect Visibility Assessment
3. Annotation Verification

Stage 1: Image Quality

- Resolution
- Blur
- Brightness
- Contrast

Stage 2: Defect Visibility

- Is the defect clearly visible?
- Is it occluded or too small?

Stage 3: Annotation Verification

- Correct class label
- polygon boundary
- Complete defect coverage

Final Pipeline

Raw Dataset → Curation → Tagging → Three-Stage Filtering → High-Quality Dataset → Instance Segmentation Annotation

Key Message

- Data quality is the foundation of model performance
- Structured curation improves annotation reliability
- Enables transition to instance-level defect detection